

The Twentieth Century as a Sequence of Ideological Phase Transitions: A Thermodynamic Reinterpretation of Modern History

Abstract

The twentieth century is usually interpreted as a struggle between ideologies: liberalism, communism, fascism, and democracy. This paper proposes an alternative interpretation: the twentieth century was not a battle of ideas, but a sequence of thermodynamic phase transitions in global social systems driven by energy regimes, entropy accumulation, and temporal irreversibility. Ideologies functioned as metastable narrative containers that temporarily stabilized civilizations under changing physical constraints. Their collapse followed not from intellectual failure, but from thermodynamic exhaustion.

1. The End of Classical Liberal Phase (1914–1930)

The 19th-century liberal order operated in a **low-entropy, high-expansion regime** fueled by:

- Colonial energy extraction
- Industrial throughput
- Demographic youth
- Financial unrealized value

This created a stable liberal metastable phase:

$$E_{high}, \quad S_{low}, \quad T_{illusion} \gg 0$$

World War I represented not a political accident but a **thermodynamic rupture**: energy and demographic losses exceeded narrative absorptive capacity.

The liberal phase did not collapse because it was unjust.

It collapsed because its entropy budget was exhausted.

2. The Interwar Instability: Search for New Phase Containers

Between 1918 and 1939, societies entered a **supercooled ideological state**:

- Liberalism had lost stability.
- No new phase had yet fully nucleated.

This allowed the emergence of:

- Fascism (entropy suppression via coercion)
- Communism (entropy denial via moral symmetry)
- Keynesianism (entropy redistribution through time)

These were not choices.

They were **phase nucleation attempts**.

3. Fascism as Entropy Export Phase

Fascism stabilized internally by exporting entropy externally.

$$S_{internal} \downarrow, \quad S_{external} \uparrow$$

War was not policy.

War was thermodynamic necessity.

Its collapse was guaranteed because entropy cannot be eliminated, only displaced.

4. Communism as Loss-Forbidden Phase

Communism attempted to construct a system where loss was ideologically illegal.

$$L \rightarrow 0$$

But loss is a physical invariant.

The result was throughput collapse and entropy paralysis.

Communism did not fail because of corruption.

It failed because it violated thermodynamic loss necessity.

5. Keynesianism as Temporal Buffer Phase

Post-WWII Western economies entered a Keynesian phase:

$$L_{socialized}, \quad T_{buffered}$$

Loss was postponed into public debt and future generations.

This created the illusion of reversible stability.

The system did not solve entropy.

It deferred it.

6. The Cold War as Phase Boundary Stabilization

The Cold War was not ideological opposition.

It was a **phase boundary stabilization** between two incompatible thermodynamic regimes.

Both systems required the other to define their metastability.

Collapse of one guaranteed destabilization of the other.

7. Neoliberal Phase: Unrealized Loss Explosion

From 1980 onward, neoliberalism re-expanded unrealized loss:

$$T_{illusion} \uparrow, \quad R_{suppressed}$$

Financialization replaced material throughput.

Entropy was monetized and circulated.

This produced the largest oniric value regime in history.

8. 2008 as Realizability Singularity

The 2008 crisis was not financial.

It was temporal.

Realizability density exceeded postponement capacity.

The system did not lack money.

It lacked future.

9. Populism as Narrative Compression Phase

Post-2008 populism emerged as:

$$N \gg E$$

Narratives attempted to compensate for thermodynamic insolvency.

This is structurally unstable.

10. The 21st Century: Post-Ideological Transition

We are now entering a regime where:

- No ideology can claim reversibility.
- No narrative can absorb entropy.
- No political system can deny loss.

The next phase will not be ideological.

It will be **thermodynamic**.

11. The Century Reinterpreted

The twentieth century was not:

- Capitalism vs socialism.
- Democracy vs totalitarianism.

It was:

A planetary system searching for a stable thermodynamic phase under irreversible time.

12. Final Statement

*Ideologies did not shape the twentieth century.
The twentieth century exhausted ideologies.*